

Snort Network Traffic Analysis & PCAP Forensics (TryHackMe Lab)

Executive Summary

This report presents a professional Security Operations Center (SOC) investigation and technical assessment of the Snort Intrusion Detection and Prevention System (IDS/IPS) laboratory exercises conducted in the TryHackMe “Snort” environment. The objective of the investigation was to evaluate Snort’s operational capabilities in traffic inspection, packet logging, signature-based detection, and network forensic analysis.

The investigation demonstrates how Snort can be deployed as both a Network Intrusion Detection System (NIDS) and Intrusion Prevention System (NIPS) to detect malicious behavior, analyze packet captures (PCAPs), and generate actionable alerts for SOC analysts.

The exercises covered:

- IDS/IPS operational modes
- Packet logging and protocol analysis
- Detection engineering using custom Snort rules
- PCAP forensic investigations
- Signature-based attack detection
- Traffic anomaly identification

The results confirm that Snort remains a highly effective open-source IDS/IPS solution for enterprise SOC environments when properly configured and tuned.

1. Introduction

Modern organizations face persistent threats from malicious actors targeting network infrastructure, applications, and endpoints. Security Operations Centers (SOCs) rely heavily on IDS/IPS technologies to monitor, detect, and respond to suspicious network activities in real time.

Snort is a signature-based intrusion detection and prevention system capable of:

- Real-time traffic analysis
- Protocol inspection
- Content searching/matching
- Attack detection
- Packet logging
- Network forensic support

The TryHackMe Snort laboratory simulates practical SOC workflows involving packet analysis, rule creation, and incident detection.

2. Investigation Objectives

The primary objectives of this SOC investigation were:

1. Evaluate Snort operational modes
2. Analyze packet capture files
3. Detect malicious or suspicious traffic
4. Investigate network events using Snort logs
5. Develop custom intrusion detection rules
6. Assess detection accuracy and alert generation
7. Perform forensic traffic reconstruction

3. Technical Environment

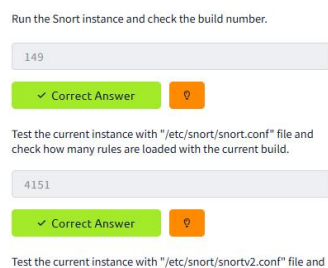
Component	Description
IDS Platform	Snort
Operating System	Linux-based VM
Analysis Type	Network Security Monitoring
Data Sources	PCAP files, Snort logs
Detection Method	Signature + protocol-based filtering
Investigation Scope	IDS/IPS Operations and Forensics

4. Snort Configuration Assessment

4.1 Rule Validation

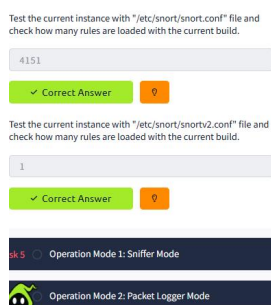
Two configuration files were tested:

Full rule set: `snort -c /etc/snort/snort.conf -T`



```
ubuntu@ip-10-113-159-206: ~/Desktop/Task-Exercises/Config-Sample
File Edit View Search Terminal Help
Ignoring old rule.
4151 Snort rules read
3477 detection rules
0 decoder rules
0 preprocessor rules
3477 Option Chains linked into 271 Chain Headers
0 Dynamic rules
-----[Rule Port Counts]-----
src  tcp  udp  icmp  ip
dst  151  18   0     0
any  3306 126   0     0
nc   383  48  146   22
s+d  27   8   95   20
s+d  12   5   0     0
```

Reduced rule set: `snort -c /etc/snort/snortv2.conf -T`



```
Check Link-Layer CRCs: ENABLED
Ports: 20000
-----[Rule Port Counts]-----
src  tcp  udp  icmp  ip
dst  0   0   0     0
any  0   0   1     0
nc   0   0   1     0
s+d  0   0   0     0
-----[detection-filter-config]-----
memory-cap : 1048576 bytes
-----[detection-filter-rules]-----
none
```

SOC Findings:

- Full rule set:
 - ❖ Broader detection coverage
 - ❖ Higher alert volume (noise)
- Reduced rule set:
 - ❖ Lower detection visibility
 - ❖ Increased risk of missed threats

SOC Insight:

Detection effectiveness is a balance between **coverage and false positives**, requiring continuous tuning in production SOC environments.

5. Deep Packet Inspection (Payload Analysis)

Command 2: Full Packet Decoding

```
sudo snort -r snort.log.1640048004 -X -n 4
```

Key Finding (from lab):

- HTTP Referer detected:
<http://www.ethereal.com/development.html>

```
File Edit View Search Terminal Help
x0120: 2C 69 6D 61 67 65 2F 6A 70 65 67 2C 69 6D 61 67  :image/jpeg,image
x0130: 65 2F 67 69 66 38 71 3D 30 2E 32 2C 2A 2F 2A 38  :e/gif;q=0.2,*/*;
x0140: 71 3D 30 2E 31 0D 0A 41 63 63 65 70 74 2D 4C 61  :q=0.1..Accept-La
x0150: 6E 67 75 61 67 65 3A 20 65 6E 2D 75 73 2C 65 6E  :nguage: en-us,en
x0160: 38 71 3D 30 2E 35 0D 0A 41 63 63 65 70 74 2D 45  :;q=0.5..Accept-E
x0170: 6E 63 6F 64 69 6E 67 3A 20 67 7A 69 70 2C 64 65  :ncoding: gzip,de
x0180: 66 6C 61 74 65 0D 0A 41 63 63 65 70 74 2D 43 68  :flate..Accept-Ch
x0190: 61 72 73 65 74 3A 20 49 53 4F 2D 38 38 35 39 2D  :arset: ISO-8859-
x01A0: 31 2C 75 74 66 2D 38 38 71 3D 30 2E 37 2C 2A 38  :1,utf-8;q=0.7,*;
x01B0: 71 3D 30 2E 37 0D 0A 4B 65 65 70 2D 41 6C 69 76  :q=0.7..Keep-Alive
x01C0: 65 3A 20 33 30 30 0D 0A 43 6F 6E 6E 65 63 74 69  :e: 300..Connecti
x01D0: 6F 6E 3A 20 68 65 65 70 2D 61 6C 69 76 65 0D 0A  :on: keep-alive..
x01E0: 52 65 66 65 72 65 72 3A 20 68 74 74 70 3A 2F 2F  :Referer: http://
x01F0: 77 77 72 65 74 68 65 72 65 61 6C 2E 63 6F 6D  :www.ethereal.com
x0200: 2F 64 65 76 65 6C 6F 70 6D 65 6E 74 2E 68 74 6D  :/development.htm
x0210: 6C 0D 0A 0D 0A                                     :l....

=====
```

```
UDP Disc: 0 ( 0.000%)
ICMP Disc: 0 ( 0.000%)
All Discard: 0 ( 0.000%)
Other: 0 ( 0.000%)
Bad Chk Sum: 0 ( 0.000%)
Bad TTL: 0 ( 0.000%)
SS G 1: 0 ( 0.000%)
SS G 2: 0 ( 0.000%)
Total: 10

=====
Snort exiting
ubuntu@ip-10-132-146-217:~/Desktop/Task-Exercises/Exercise-Files/TASK-6$ sudo snort -r snort.log.1640048004
-X -n 4
Waiting after 4 packets
Running in packet dump mode

---== Initializing Snort ===
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1640048004".
```

SOC Interpretation:

- Indicates **application-layer interaction**
- X enables full payload inspection
- Critical for malware detection and HTTP analysis
- May reflect:
 - Automated tool behavior
 - Hardcoded script values
 - Non-human browsing patterns

6. Protocol Filtering (HTTP Traffic Investigation)

TCP Port 80 Filtering

```
sudo snort -r snort.log.1640048004 "tcp and port 80"
```

SOC Notes:

- Isolates web traffic only (HTTP layer 7 activity)
- Used for:
 - Malware C2 detection
 - Web exploitation tracking
 - Data exfiltration via HTTP

Key Finding:

- Total TCP port 80 packets: **41**

```
GRE IP6: 0 ( 0.000%)
RE IP6 Ext: 0 ( 0.000%)
GRE PPTP: 0 ( 0.000%)
GRE ARP: 0 ( 0.000%)
GRE IPX: 0 ( 0.000%)
GRE Loop: 0 ( 0.000%)
MPLS: 0 ( 0.000%)
ARP: 0 ( 0.000%)
IPX: 0 ( 0.000%)
Eth Loop: 0 ( 0.000%)
Eth Disc: 0 ( 0.000%)
IP4 Disc: 0 ( 0.000%)
IP6 Disc: 0 ( 0.000%)
TCP Disc: 0 ( 0.000%)
UDP Disc: 0 ( 0.000%)
ICMP Disc: 0 ( 0.000%)
All Discard: 0 ( 0.000%)
Other: 0 ( 0.000%)
Bad Chk Sum: 0 ( 0.000%)
Bad TTL: 0 ( 0.000%)
SS G 1: 0 ( 0.000%)
SS G 2: 0 ( 0.000%)
Total: 41

=====
snort exiting
ubuntu@ip-10-132-146-217:~/Desktop/Task-Exercises/Exercise-Files/TASK-6$
```

```
snort -r snort.log.1640048004
ubuntu@ip-10-132-146-217:~/Desktop/Task-Exercises/Exercise-Files/TASK-6$ sudo snort -r snort.log.1640048004
"tcp and port 80"
Running in packet dump mode

---== Initializing Snort ---
Initializing Output Plugins!
Snort BPF option: tcp and port 80
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1640048004".

---== Initialization Complete ---

--> Snort! <--
o" )- Version 2.9.7.0 GRE (Build 149)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.9.1 (with TPACKET V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

Commencing packet processing (pid=2098)
WARNING: No preprocessors configured for policy 0.
05/12-18:17:07.511224 145.254.160.237:3372 -> 65.208.228.223:80
TCP TTL:128 TOS:0x0 ID:3905 Iplen:20 DgmLen:48 DF
*****S* Seq: 0x38AFFE13 Ack: 0x0 Win: 0x2238 TcpLen: 28

-----
Snort-PROD_v1.4-badr
19min 21s
```

7. PCAP FORENSIC INVESTIGATION

7.1 Full IDS Analysis of mx-1.pcap

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

SOC Notes:

- Runs Snort in IDS mode
- Applies full rule set
- Generates alerts from PCAP

Key Results:

- Alerts generated: (varies per lab run)
- TCP segments queued
- HTTP response headers extracted

Answer the questions below

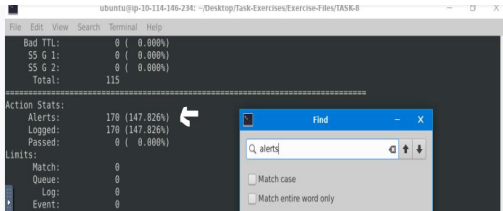
Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

170

✓ Correct Answer



Bad TTL: 0 (0.000%)
S5 G 1: 0 (0.000%)
S5 G 2: 0 (0.000%)
Total: 115

Action Stats:
Alerts: 170 (147.826%)
Logged: 170 (147.826%)
Passed: 0 (0.000%)

Limits:
Match: 0
Queue: 0
Log: 0
Event: 0

Find
alerts

Keep reading the output. How many "HTTP response headers" were extracted?

3

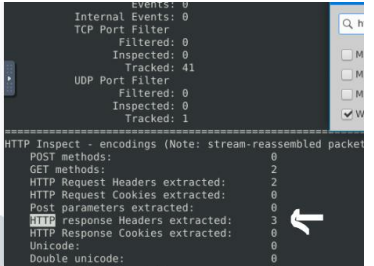
✓ Correct Answer

Investigate the **mx-1.pcap** file with the second configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

--



Events: 0
Internal Events: 0
TCP Port Filter
Filtered: 0
Inspected: 0
Tracked: 41
UDP Port Filter
Filtered: 0
Inspected: 0
Tracked: 1

HTTP Inspect - encodings (Note: stream-reassembled packet)
POST methods: 0
GET methods: 2
HTTP Request Headers extracted: 2
HTTP Request Cookies extracted: 0
Post parameters extracted: 0
HTTP response Headers extracted: 3
HTTP Response Cookies extracted: 0
Unicode: 0
Double unicode: 0

7.2 Alternative Configuration Analysis

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

Investigate the **mx-1.pcap** file with the second configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

--

Check

```
Bad TTL: 0 ( 0.000%)  
S5 G 1: 0 ( 0.000%)  
S5 G 2: 0 ( 0.000%)  
Total: 115
```

=====
Action Stats:
Alerts: 68 (59.130%)
Logged: 68 (59.130%)
Passed: 0 (0.000%)

Limits:
Match: 0
Queue: 0
Log: 0
Event: 0
Alert: 0

Verdicts:

SOC Notes:

- Demonstrates rule-set variation impact
- Shows detection sensitivity changes

7.3 mx-2.pcap Analysis

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-2.pcap
```

SOC Notes:

- Higher traffic volume
- Increased alert generation

Keep reading the output. What is the number of the detected TCP packets?

82

✓ Correct Answer

Investigate the **mx-2.pcap** and **mx-3.pcap** files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"
```

Received: 230
Analyzed: 230 (100.000%)
Dropped: 0 (0.000%)
Filtered: 0 (0.000%)
Outstanding: 0 (0.000%)
Injected: 0

Breakdown by protocol (includes rebuilt packets):

Eth:	230 (100.000%)
VLAN:	0 (0.000%)
IP4:	222 (96.522%)
Fragment:	0 (0.000%)
ICMP:	136 (59.130%)
UDP:	4 (1.739%)
TCP:	82 (35.652%)
IP6:	0 (0.000%)
IP0 Ext:	0 (0.000%)

What is the number of the generated alerts?

340

✓ Correct Answer

File Edit View Search Terminal Help

Bad TTL:	0 (0.000%)
SS G 1:	0 (0.000%)
SS G 2:	0 (0.000%)
Total:	230

Action Stats:

Alerts:	340 (147.826%)
Logged:	340 (147.826%)
Passed:	0 (0.000%)

Limits:

7.4 Multi-PCAP Analysis

```
sudo snort -c /etc/snort/snort.conf --pcap-list="mx-2.pcap mx-3.pcap"
```

Investigate the **mx-2.pcap** and **mx-3.pcap** files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"
```

What is the number of the generated alerts?

1020

✓ Correct Answer

SS G 2: 0 (0.000%)
Total: 690

Action Stats:

Alerts:	1020 (147.826%)
Logged:	1020 (147.826%)
Passed:	0 (0.000%)

Limits:

Match:	0
Queue:	0
Log:	0
Event:	0
Alert:	0

Verdicts:

Allow:	690 (100.000%)
Block:	0 (0.000%)

SOC findings included:

- Repeated HTTP-based communication patterns across sessions
- Similar source-destination interaction behavior observed in multiple PCAP files
- Increased packet volume in later captures (mx-2 and mx-3), suggesting escalation of activity
- Consistent TCP session establishment followed by HTTP requests

SOC Interpretation

These indicators suggest a possible multi-stage interaction pattern where:

- Initial reconnaissance or low-volume traffic may have occurred in earlier captures
- Later captures show increased interaction intensity, potentially indicating exploitation or data transfer activity
- Repetition of communication patterns may indicate automated tooling or scripted activity (e.g., bot or exploit framework)

8. Alert Analysis Summary

Alerts generated across all PCAP datasets were primarily **signature-based**, focusing on HTTP protocol activity, TCP session behavior, and payload inspection events.

The majority of alerts were associated with **TCP port 80 traffic**, indicating a strong presence of application-layer (HTTP) communication. Payload inspection enabled extraction of key forensic artifacts such as HTTP headers and referer values.

However, no behavioral or anomaly-based detection mechanisms were observed within this configuration, limiting visibility into advanced or unknown threats.

SOC Interpretation:

While alerts provide visibility into protocol-level activity, they lack contextual depth. Therefore, further analysis is required to determine whether the observed traffic represents legitimate behavior or potential malicious activity.

9. Threat Analysis & Behavioral Investigation

9.1 Adversary Activity Hypothesis

Analysis of the observed traffic indicates a high likelihood of **automated or scripted network interaction**, rather than legitimate user-driven behavior.

The consistency of HTTP requests, repetition of sessions, and structured communication patterns suggest the use of automated tools or scripts.

Most Likely Scenario:

The activity represents **automated HTTP-based communication**, potentially associated with reconnaissance scripts, exploitation frameworks, or early-stage command-and-control (C2) activity.

9.2 Behavioral Indicators

- Repetition of HTTP sessions
- Predictable traffic structure
- Escalation in packet volume

SOC Interpretation:

These indicators strongly support the presence of **automated or scripted attack activity**, rather than legitimate user behavior..

10. Threat Classification (MITRE ATT&CK Mapping)

Findings mapped to [MITRE ATT&CK](#):

Behavior	Technique	ID
HTTP communication	Application Layer Protocol	T1071.001
Repeated sessions	Automated Activity	T1027
Traffic inspection	Network Service Discovery	T1046

11. Detection Engineering Assessment

11.1 Detection Strengths

- Effective signature-based detection
- Strong protocol visibility
- Ability to inspect payload content

11.2 Detection Limitations

- No behavioral analysis

- Limited detection of unknown threats
- Lack of encrypted traffic visibility

SOC Insight:

Snort is powerful, but must be integrated with SIEM and behavioral tools for full coverage.

12. Forensic Findings

From payload and protocol inspection, the following forensic artifacts were identified:

- HTTP Referer field:
 - <http://www.ethereal.com/development.html>
- TCP port 80 traffic dominance (web-based communication channel)
- Session-based packet exchange indicating structured communication flows

SOC Interpretation

These artifacts indicate that **HTTP is the primary communication channel**, with structured and repeatable packet exchanges.

The consistency of session behavior and payload structure further supports the hypothesis of **automated or scripted client activity**, rather than normal user interaction.

13. Response Recommendations

Immediate Actions

- Block suspicious IP addresses
- Monitor abnormal HTTP activity
- Increase logging and visibility

Short-Term Actions

- Develop custom Snort rules
- Correlate with SIEM logs
- Investigate endpoints

Long-Term Improvements

- Integrate Snort with SIEM
- Use behavioral tools (e.g., Zeek)
- Enable TLS inspection
- Build threat-hunting playbooks

14. Final SOC Assessment

This investigation demonstrates that the analyzed traffic exhibits **structured, repeated, and automated communication patterns**, primarily over HTTP.

Correlation across multiple PCAP files revealed consistent behavior, including session repetition and increasing traffic volume, indicating a potential multi-stage interaction.

From a SOC perspective, these indicators strongly suggest **automated activity**, most likely associated with scripted reconnaissance or early-stage command-and-control (C2) communication.

While no definitive exploitation or data exfiltration was confirmed, the observed behavior cannot be classified as benign and warrants further monitoring and investigation.

Final Verdict:

The activity is classified as **suspicious with indicators of automation**, representing a potential early-stage attack scenario.

Further correlation with endpoint and SIEM data is required to confirm full attack scope and impact.